

NAME: 拼音 _____ 汉字 _____

Checkpoint 2 Mock Test [94 marks]

1. **Draw** the Born-Harbor cycle of MgCl_2 .

For each step, **indicate** the **name** of the corresponding enthalpy and **label** the reactant(s)/product(s). **Search** the numerical value of each enthalpy and **calculate** the lattice energy of MgCl_2 .

[12 marks in total]

2. Acyclic alkane **A** is chiral. Isotope tritium is not present in **A**. [12 marks in total]

a) If **A1** contains C, H and D. What is the minimum number of carbon atoms in **A1**?

Draw the skeletal structure of the (R)-isomer. [3 marks]

b) If **A2** contains C and H, but no D. What is the minimum number of carbon atoms in **A2**? **Draw** the skeletal structure of the (S)-isomer. [3 marks]

- c) How many mono-substituted product(s) can be formed, when (R)-**A1** is reacted with chlorine gas under suitable condition? **Draw** their skeletal structures. [6 marks]

3. pH control is important not only in chemistry, but also in biology, material science, environmental science and engineering. [12 marks in total]

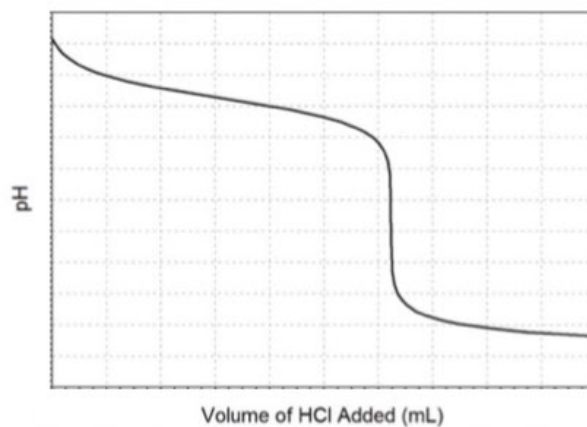
a) **Calculate** the pH of $0.100 \text{ mol dm}^{-3}$ HCl solution [1 mark]

b) **Calculate** the pH of $0.100 \text{ mol dm}^{-3}$ NH_3 solution ($\text{pK}_b = 4.75$). [2 marks]

10.0 cm^3 of solution b) is titrated with solution a). The pH value is continuously measured and the titration curve is plotted.

c) What is the volume (V_1) of solution a), when end point is reached? [1 mark]

d) Calculate the pH of the end point. [3 marks]



- e) Calculate the pH of the mixture, when the volume of the titrant added $V_2 = \frac{2}{3}V_1$.
[5 marks]

4. Synthesis of ammonia is crucial to human survival because it is a key component in fertilizers, which are essential for growing the food that sustains the global population. [22 marks in total]

- a) **Write** the equation for the synthesis of ammonia with state symbols. [2 marks]

- b) **Calculate** the $\Delta_r H^\circ$, $\Delta_r S^\circ$ and $\Delta_r G^\circ$ of the reaction at 350 °C. Assume the change of enthalpy and entropy are temperature-independent. [6 marks]

- c) **Calculate** the equilibrium constant K of the reaction. [2 marks]

A reaction vessel is kept at constant pressure of 200 bar and constant temperature of 350 °C. The mole fraction of gases inside the vessel is 20% NH_3 , 20% N_2 and 60% H_2 .

- d) **Calculate** Q of the reaction and **determine** the direction of the reaction. [3 marks]

- e) When the reaction reaches equilibrium at 350 °C and 200 bar, **calculate** the partial pressure of each component. [9 marks]

5. Hess's Law is useful in calculation of enthalpy change of reactions, particularly the reactions that are too dangerous to be experimentally conducted.

Using Hess's Law, **calculate** $\Delta_r H^\circ$ of the reaction between liquid methylhydrazine ($\text{CH}_3\text{-NH-NH}_2$) and dinitrogen tetroxide. The products are nitrogen, carbon dioxide and liquid water. [12 marks]

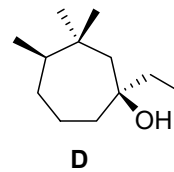
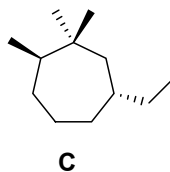
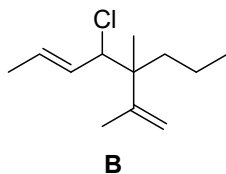
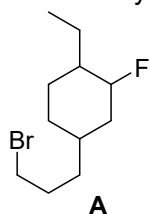
You may choose to use the following data:

Chemical	N_2O_4 (g)	CO_2 (g)	CO (g)	H_2O (l)	H_2O (g)
$\Delta_f H^\circ$ (kJ mol ⁻¹)	9.08	-393.52	-110.53	-285.83	-241.82

Enthalpy of combustion of methylhydrazine (l): $-1305.2 \text{ kJ mol}^{-1}$ (298 K)

Enthalpy of combustion of carbon monoxide: $-283.0 \text{ kJ mol}^{-1}$ (298 K)

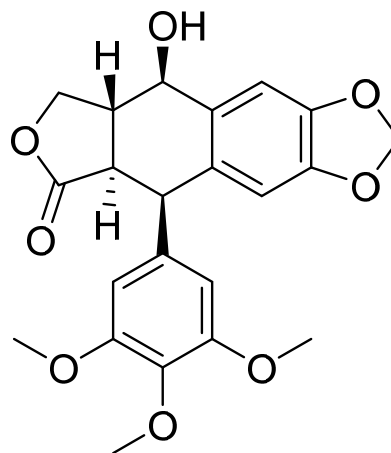
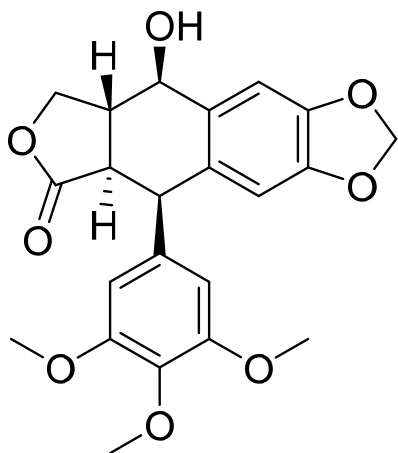
6. **Write** the IUPAC name of the following compounds, paying special attention to stereochemistry when applicable. [12 marks]



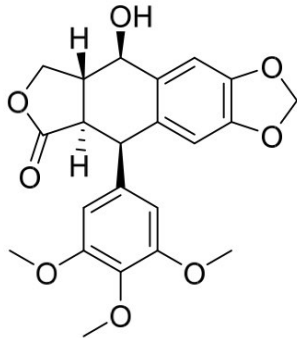
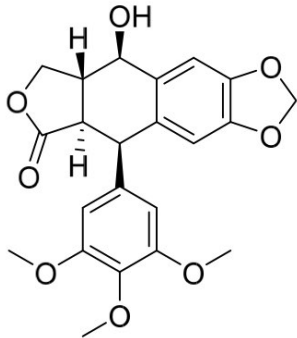
A	
B	
C	
D	

[Hint: when there is a functional group cited as suffix, e.g. -ol in cycloheptanol, the locant number of -OH group should be as low as possible. This rule takes higher priority than other rules about locant number. This hint will NOT be included in the actual test.]

7. For each chiral centre in the given molecule, **rank** the groups by their priority and **assign** the absolute configuration. If you need to label two chiral centers in one graph, choose two chiral centers far away and use two different colors. [12 marks]



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Simplified IUPAC rule of nomenclature for organic compounds

1. Maximum number of principle group cited as suffix
2. Maximum number of C=C and C≡C bonds together
3. Maximum length
4. Maximum number of C=C bonds
5. Lowest locant(s) for principle group(s)
6. Lowest locant(s) for multiple bond(s)
7. Lowest locant(s) for double bond(s)
8. Maximum number of substituents cited as prefix
9. Lowest locant(s) for the substituent(s) cited as prefix
10. Lowest locant for the first cited substituent

Note 1: This is the “full” list of IUPAC nomenclature rule (for simple molecules).
The rules in red has been taught in the lecture.

Note 2: Regarding the rules involving principle functional groups (Rule 1 and 5),
only alcohol functional group will be tested (see hint of Question 6).

Note 3: Halo groups (fluoro, chloro, bromo, iodo) are always treated as prefix.
Alkene (-ene) and alkyne (-yne) are part of the parent chain/ring, and
considered neither as prefix nor as suffix.

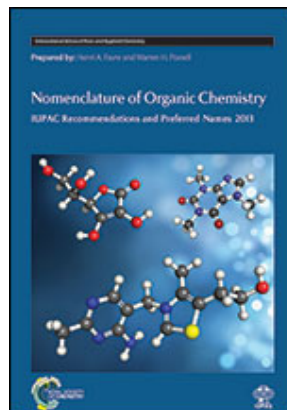
Note 4: C≡C will not be tested.

Useful resources for nomenclature (you may need them in the future):

1. **Nomenclature of Organic Chemistry: IUPAC Recommendations and Preferred Names 2013**, IUPAC Blue book, prepared for publication by Henri A Favre and Warren H Powell, by RSC Publishing, 2014 [ISBN 978-0-85404-182-4]; <https://doi.org/10.1039/9781849733069>

2. **Nomenclature of Organic Compounds**, by S.C. Pal, Oxford: Alpha Science, 2006.

3. https://en.wikipedia.org/wiki/IUPAC_nomenclature_of_organic_chemistry



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